

ANIL PRADHAN

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AI & Backend Engineer specializing in autonomous multi-agent architectures, computer vision pipelines, and resilient backend systems with Python, FastAPI, LangGraph, OpenCV, and YOLO11n.

Dell FutureMinds National Grand Finalist (Top 6 / 8,000+) · 4× AI Hackathon Builder (Dell, MSFT, AWS, Google)

EDUCATION

Institute of Technical Education & Research (ITER), SOA University

Bachelor of Technology (B.Tech), Computer Science & Engineering

2024 – 2028

CGPA: 8.93

Nalanda Vidya Mandir Higher Secondary School, Berhampur, Odisha

Class XII (Higher Secondary) — Physics, Chemistry, Mathematics with IT

2022 – 2024

Score: 85%

Odisha Adarsha Vidyalaya (OAV), Digapahandi, Ganjam, Odisha

Class X (CBSE Board)

2020 – 2022

Score: 78.8%

TECHNICAL PROJECTS

VisionForge AI

Aug 2026 – Present

[GitHub](#) — AI/ML & Pipeline Engineer | Team: Disha & Anil

- Built an autonomous inspection pipeline using **LangGraph state machines** to run OCR, seal-matching, and YOLO11n detection in parallel, with **Groq and Gemini Vision** models giving the final pass/fail verdict in under 5s.
- Trained and fine-tuned a custom **YOLO11n model on 4,448 hardware images** to accurately detect and locate electronic components as small as **12×12 px**.
- Developed an anomaly detection approach (**Anomaly Max-Pooling**) using **OCR and FAISS** to improve detection of critical hardware defects, with **203 automated tests**.
- Implemented **real-time inspection updates using Server-Sent Events (SSE)**, allowing operators to track pipeline progress and receive inspection results instantly.

Tech: Python · FastAPI · LangGraph · YOLO11n · OpenCV · FAISS · Groq LPU · Gemini 3.5 Flash

TaskPilot AI

Jul 2026 – Present

[Live Demo](#) | [GitHub](#) — Integration Lead | Team IdeaForg-E

- Built the **orchestrator** for a 6-stage AI pipeline, coordinating task collection, extraction, deduplication, quality checks, prioritization, and scheduling across **5 data sources**.
- Implemented the **pipeline lifecycle and state management** to handle stage execution and recovery, completing the full workflow in around **14–18 seconds**.
- Built a **reliable LLM integration** with fallback mechanisms, allowing the system to continue working during API failures and rate limits.
- Implemented **real-time P1 task injection** through the AI copilot, automatically triggering a pipeline re-run and moving urgent incidents to the **#1 priority**.

Tech: Python · FastAPI · Multi-Agent Systems · SQLAlchemy · SQLite (WAL) · Groq · React 18

AI Career Mentor

Feb 2026 – Aug 2026

[Live Demo](#) | [GitHub](#) — Solo Developer

- Built a multi-agent AI system using **LangGraph** to orchestrate 4 parallel agents (Resume, Market, Roadmap, LinkedIn) with shared state, reducing analysis latency by **~60%**.
- Developed a real-time mock interview engine with a **7-phase FSM over WebSockets**, featuring a live code editor and resume-personalized questions.
- Designed a fault-tolerant backend (**FastAPI, PostgreSQL, Redis**) with **RAG-powered roadmap generation via ChromaDB**, multi-provider LLM failover, and a **113-test CI/CD pipeline**.

Tech: FastAPI · LangGraph · Next.js · ChromaDB · Redis · Docker · ONNX

TRAINING

Agentic AI Training Program — CogniForce.AI & ITER, SOA University

2026 – Present

- Hands-on training in **Agentic AI, LLMs, prompt engineering, RAG, tool calling, MCP**, and multi-agent orchestration.
- Built and experimented with autonomous agents using **OpenClaw, Multica, AnythingLLM, OpenRouter, Claude Code**, and custom MCP servers.

TECHNICAL SKILLS

Languages: Python, Java, SQL

AI & Computer Vision: LangGraph, Multi-Agent Systems, RAG, Prompt Engineering, LLM Integration, OpenCV, YOLO11n, OCR, Anomaly Detection (SSIM)

Backend & Databases: FastAPI, REST APIs, SQLAlchemy, JWT/RBAC, PostgreSQL, Redis, ChromaDB, SQLite

Core CS Concepts: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

Developer Tools: Git/GitHub, Docker, Linux Commands, Postman

ACHIEVEMENTS & CERTIFICATIONS

- Dell FutureMinds AI Hackathon 2026 — National Grand Finalist:** Top 6 nationwide out of 6,000+ participants; pitched our live solution at Dell Bengaluru Office.
- FlowFinance × Techfest IIT Bombay — Round 2:** Developed GTM strategy and acquisition framework for AI fintech platform.
- 4× AI Hackathon Builder:** Deployed AI applications in Microsoft AI Dev Days, Amazon Nova, and Google Solution Challenge.
- 100+ Day DSA Streak:** Continuous daily problem solving on GeeksforGeeks 160 Days of Code.
- Certifications:** IBM AI Fundamentals · Cisco Python 2 · Wells Fargo SWE · HackerRank (DSA, SQL, Java, Python).